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A continuous-time Markov chain and stochastic differential equations approach for modeling malaria propagation

Asma Akter Akhi [a], Md. Kamrujjaman [a,*], Kaniz Fatema Nipa [b], Taufiquar Khan [c]

[a] Department of Mathematics, University of Dhaka, Dhaka 1000, Bangladesh
[b] Odum School of Ecology, University of Georgia, Athens, GA 30602, USA
[c] Department of Mathematics and Statistics, University of North Carolina at Charlotte, Charlotte, NC, USA

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ABSTRACT

Malaria is the world's most fatal and challenging parasitic disease, caused by the Plasmodium parasite and transmitted to humans by the bites of infected female mosquitos. This study presents a stochastic process representing the evolution in time of an unexpected phenomenon. We use different probability techniques to assume the possible outcome in a Malaria model since state variables or parameters are random in a stochastic model. This study considers a deterministic vector-host malaria model and converts this model into the corresponding stochastic model with a Continuous-time Markov chain (CTMC) and Stochastic differential equation (SDE). We prove the positivity and boundedness of solutions and calculate the equilibrium points. The threshold values are evaluated using different approaches. Moreover, the Latin Hypercube Sampling (LHS) and Partial Rank Correlation Coefficients (PRCC) procedures are used to analyze the sensitivity of each model parameter. Finally, a transient numerical simulation is discussed, and their outcomes strongly correlate with the actual scenario.

1. Introduction

Vector-borne diseases have been caused by transmitting the pathogenic microorganism from an infected subjective to another subjective by different types of arthropods or parasites like mosquitoes, kissing bugs, tsetse flies, lice, sand flies, ticks, and so on. Malaria, avian malaria, dengue, yellow fever, zika, chykungunya, etc. are known as mosquito-borne diseases where humans are the most common host [1]. Malaria is one of the world's most complex and lethal parasitic infections. Malaria is precipitated by the Plasmodium parasite and is circulated to humans through the bites of infected female Anopheles mosquitoes. Female mosquitoes need blood meals to produce eggs, which serve as the intermediary between the human (host) and the insect (vector) to complete a cycle (Fig. 1).

Malaria has become a prevalent human infection throughout the world, and all most half of the population in the world is in danger because of malaria. According to statistics from the World Health Organization (WHO), there were an estimated 228 million new cases of malaria in 2018 (range: 206–258 million), leading to approximately 405 thousand disease-related fatalities (World Health Organization 2019). Moreover, sub-Saharan Africa remains the most unresistant region with a high rate of deaths because of malaria. Bangladesh, a country with a population of more than 160 million, is similarly known as a malaria-endemic region, with 17.5 million people at risk of contracting the disease [2]. Moreover, 12–26% of *P. falciparum* malaria victims in Bangladesh were at elevated risk for *P. vivax* parasitemia, corresponding to a meta-synthesis of patient records from three sites [3]. Nowadays, the incidence of malaria has been revolting due to drug resistance.

Mbogo et al. have used a deterministic model for malaria transmission with relapse. They compared the behavior of disease dynamics for stochastic and deterministic models based on randomness. They claimed that initial conditions might affect stochastic thresholds while deterministic thresholds could not [4]. For a single group of infections, the probability of a significant outbreak of a disease is about $1-\left(\frac{1}{\mathfrak{R}_0}\right)^i$, where i be infectious individuals and $\mathfrak{R}_0 > 1$ is the basic reproduction number. When there are several infectious subgroups, stochastic thresholds rely on the population size and the probability of disease extinction within each group [5]. Allen introduced the topic of stochastic epidemic modeling concerning the concept of ODE epidemical dynamics. She was particularly prominent in continuous-time Markov chain and stochastic differential equations [6]. Nowadays, mosquito-borne diseases are a global burden, and we need to find leading indicators to eliminate mosquito-borne diseases. Using bed nets, reducing human infection period, shortening adult mosquito lifespan, and closing breeding sites can

* Corresponding author.
E-mail addresses: akhiasma752@gmail.com (A.A. Akhi), kamrujjaman@du.ac.bd (M. Kamrujjaman), KanizFatema.Nipa@uga.edu (K.F. Nipa), taufiquar.khan@uncc.edu (T. Khan).

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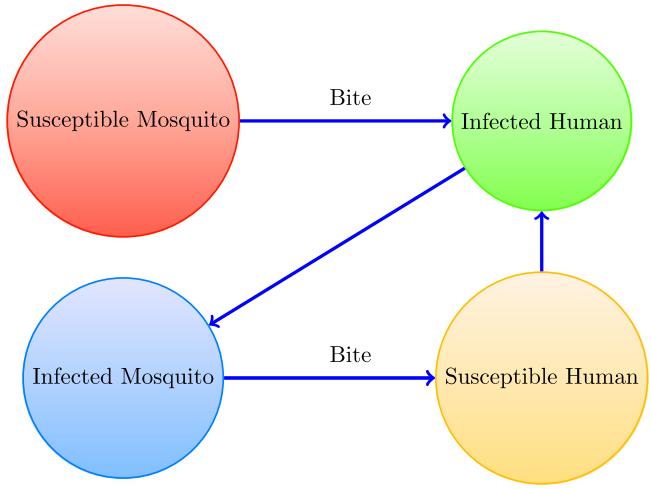


Fig. 1. The mosquito-human transmission cycle. When susceptible mosquitoes bite infected humans, they become infected via the blood of the mosquito ingests. Again, when infected mosquitoes bite susceptible human, then those humans become infected. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

be the controlling activities for eliminating the vector-borne disease. Moreover, O'Regan et al. suggested that critical slowing down was detectable in human-host fluctuations of Ross-Macdonald-type vector-borne disease systems [7]. Al-Awadhi, Ahmad, and Iqbal investigated the current status of malaria epidemiology in the middle east region and beyond. They claimed that imported malaria cases are increasing remarkably in the malaria-free country and nonendemic eras because of the changes in climate and ecosystem, increasing travel for business, migratory movements for employment, and so on [8]. The probability that a mosquito bites an infectious human will increase with the number of infected mosquitoes and decrease according to the number of infected humans. Furthermore, by improving human immunity, the number of infected mosquitoes can be declined [9]. Increasing the relapse rate will also help to eradicate malaria [10]. Particular time or seasonal variation can affect the behavior of many vector-borne diseases. Thus, the risk of a disease outbreak can be periodic and depends on the seasonal variability and the number of initially infected populations [11–15]. Estimating the probabilities of a disease outbreak from CTMC models and branching process approximation coincide when the population size is sufficiently large, and the basic reproduction number is more extensive than one [16]. The branching process approximation may not calculate a good approximation if the number of host and vector populations is less than 100 [11,16]. The probability of an outbreak is approximated by using the CTMC model for different values of parameters, and the initial number of infected individuals for COVID-19 in [17–20].

There has been an increase in malaria infections and fatalities due to the lockdown and restrictions on the movement of healthcare workers brought on by the COVID-19 pandemic. These disruptions include the distribution of insecticide-treated bed nets and seasonal malaria chemotherapy [21,22]. Efficiency and economic assessments are essential in epidemiological modeling for discovering effective and affordable intervention strategies that prevent the most significant number of infection cases at the lowest implementation cost [23–25]. The IMPACT study in Tanzania, the Uganda Malaria Monitoring Project, and the Southeast African Combined Antimalarial Treatments Project are three African programs investigating malaria prevention, health care monitoring, and policy regarding public health [26]. In order to effectively stop the spread of COVID-19, Chowdhury et al. estimated weighted transmissibility that has taken into account the percentage of current strains that naturally change over time [27].

Significant heterogeneity among humans and mosquitoes can affect the dynamics of malaria transmission. For example, human immunity, mosquito biting rates, and treatment-seeking patterns differ across individuals and populations. The mosquito biting behavior is stochastic, with the frequency of bites, preferred feeding methods, and infectivity rates. Additionally, individual variations have been observed in the dynamics of immune responses, immunity development, and parasite growth [28–30]. Moreover, interventions like vector control tactics (e.g., insecticide-treated bed nets, domestic residual spraying) and treatment regimens (e.g., medicine administration and case supervision) introduce additional sources of diversity and randomness. Thus, a stochastic model is well justified in studying malaria due to the biological underpinnings. In this study, we apply the stochastic model to examine malaria's inherent biological complexity and variability. It offers a more realistic representation of the transmission behaviors, infection results, and interventional effects. The stochastic model helps to explore the probabilistic nature of malaria and enhances a deeper understanding of malaria epidemiology, prevention methods, and potential effects of treatments in various scenarios.

The article is organized as follows: mathematical model is discussed elaborately with positivity and boundedness of solutions, disease-free equilibrium (DFE) point, and primary reproduction number in Sections 2, 2.1, A.1, and A.2. Further, the endemic equilibrium point is calculated and shows the impact of the basic reproduction number for a disease outbreak in Appendix A. Then, local and global stability analysis of DFE is presented in Appendix B. Section 3 presents sensitivity analysis using LHS and PRCC procedures. The following two sections discuss the formulation of the CTMC model and the SDE model (Sections 4, 5). The well versed outcomes are presented graphically and discussed in their interpretation in Section 6. Finally, the results are summarized in Section 7.

2. Mathematical model

In the basic malaria model, Ross considered affected and unaffected individuals for both hosts (human) and vectors (mosquito) with constant population size and took infectious periods as exponentially distributed. Next, this model is modified by Macdonald, and in this study, we consider the Ross-Macdonald model for our further investigation of malaria. Assumed that host and vector populations are homogeneously mixed in the Ross-Macdonald model [31–33].

Without changing the generality or dynamical behavior of the system, the total host populations are divided into three epidemiological compartments: susceptible (S_h), infected (I_h), and recovered (R_h). Meanwhile, female mosquitoes are separated as susceptible (S_v), and infected (I_v) individuals. That means the classical SIR epidemic process is used to model host infection; on the other hand, the vector infection is modeled by the SI epidemic process [31]. The deterministic system for malaria is given below:

$$\begin{cases} \dfrac{dS_h}{dt} = \Lambda_h - \beta_{hv} I_v \dfrac{S_h}{N_h} - \mu_h S_h, & t \geq 0, \\ \dfrac{dI_h}{dt} = \beta_{hv} I_v \dfrac{S_h}{N_h} - (\gamma_h + \mu_h) I_h, & t \geq 0, \\ \dfrac{dR_h}{dt} = \gamma_h I_h - \mu_h R_h, & t \geq 0. \end{cases} \tag{2.1}$$

$$\begin{cases} \dfrac{dS_v}{dt} = \Lambda_v - \beta_{vh} S_v \dfrac{I_h}{N_h} - \mu_v S_v, & t \geq 0, \\ \dfrac{dI_v}{dt} = \beta_{vh} S_v \dfrac{I_h}{N_h} - \mu_v I_v, & t \geq 0. \end{cases} \tag{2.2}$$

with initial conditions,

$$S_h(0) = S_{h0}, \quad I_h(0) = I_{h0}, \quad R_h(0) = R_{h0}, \quad S_v(0) = S_{v0}, \quad I_v(0) = I_{v0}, \tag{2.3}$$

and, the total population for the model is defined by,

$$\boxed{N_h(t) \equiv S_h(t) + I_h(t) + R_h(t), \qquad \text{and} \qquad N_v(t) \equiv S_v(t) + I_v(t).}$$

Table 1
Descriptions of the model parameters.

Notation	Explanation
Λ_h	Recruitment/birth rate of host
Λ_v	Recruitment/birth rate of vector
μ_h	Mortality rate of host
μ_v	Mortality rate of vector
γ_h	Host recovery rate
b	The mean rate of mosquito bites per mosquito
p_h	The probability of infection transmitted to host per bite
p_v	The probability of vector infection per bite on infectious hosts
β_{hv}	The transmission probabilities per bite from infectious vector
β_{vh}	The transmission probabilities per bite from infectious host

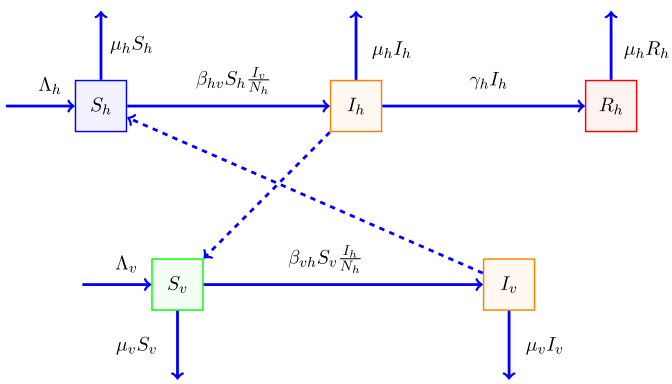


Fig. 2. Compartmental diagram for the malaria model.

Here, the total number of hosts (human) and vectors (adult mosquitoes) are N_h and N_v, respectively. Λ_h and Λ_v be the recruitment/birth rate of the hosts and vectors. μ_h and μ_v represent the number of deaths per unit of time for hosts and vectors, respectively. The host recovery rate is γ_h. Suppose, b is the number of bites per vector, i.e. the biting rate on a host. The probability of infection transmission to the host is p_h and the probability of a healthy mosquito becoming infected is p_v. So, the transmission probability from an infected vector β_{hv} can be defined as $\beta_{hv} = p_h b$. Besides, the transmission probability from an infected host β_{vh} may be decomposed as $\beta_{vh} = p_v b$. Thus, $\beta_{hv} I_v \frac{S_h}{N_h}$ is the susceptible host infection rate and $\beta_{vh} S_v \frac{I_h}{N_h}$ is the susceptible mosquito (vector) infection rate. Table 1 is presenting all model parameters. The schematic diagram for the malaria model is presented in Fig. 2.

From the model, the total population size for the hosts satisfies,

$$\frac{dN_h}{dt} = \Lambda_h - \mu_h S_h - \mu_h I_h - \mu_h R_h = \Lambda_h - \mu_h(S_h + I_h + R_h) = \Lambda_h - \mu_h N_h.$$

which is asymptotically constant, that means,

$$N_h(t) \to \frac{\Lambda_h}{\mu_h}.$$

Similarly, the total vector population size satisfies,

$$\frac{dN_v}{dt} = \Lambda_v - \mu_v S_v - \mu_v I_v = \Lambda_v - \mu_v(S_v + I_v) = \Lambda_v - \mu_v N_v.$$

which is also asymptotically constant, that means,

$$N_v(t) \to \frac{\Lambda_v}{\mu_v}.$$

In our model, we consider that recovered populations are no longer susceptible. So, if the individuals return from the disease, they will recover permanently. Thus, we can reduce the considered model into four equations by eliminating the equation for the recovery host. Then the vector-host model can be reduced to,

$$\begin{cases} \frac{dS_h}{dt} = \Lambda_h - \beta_{hv} I_v \frac{S_h}{N_h} - \mu_h S_h, & t \geq 0, \\ \frac{dI_h}{dt} = \beta_{hv} I_v \frac{S_h}{N_h} - (\gamma_h + \mu_h) I_h, & t \geq 0, \\ \frac{dS_v}{dt} = \Lambda_v - \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v S_v, & t \geq 0, \quad \text{and} \\ \frac{dI_v}{dt} = \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v I_v, & t \geq 0. \end{cases} \tag{2.4}$$

where, $N_h = \frac{\Lambda_h}{\mu_h}$ and $N_v = \frac{\Lambda_v}{\mu_v}$.

2.1. Positivity and boundedness of solutions

The positivity and boundedness of the solutions have monumental aspects in the analysis of an epidemiological model [34]. Boundedness and positivity indicate that any outcome with positive starting values will persist positively for all the non-negative variables for $t > 0$. Moreover, positivity suggests that the population will persist for an extended period. Therefore, we must establish the recommended model's positivity to check its biological validity.

Depending on the compartment system in compact form,

$$\mathbf{X}' = F(\mathbf{X}, t), \quad \mathbf{X}(0) = \mathbf{X}_0 \tag{2.5}$$

$$\mathbf{Y}' = G(\mathbf{Y}, t), \quad \mathbf{Y}(0) = \mathbf{Y}_0 \tag{2.6}$$

$\mathbf{X} = (X_1, X_2, X_3) = (S_h, I_h, R_h)$, and $\mathbf{Y} = (Y_1, Y_2) = (S_v, I_v)$. Furthermore, define $\kappa_1 = \frac{\Lambda_h}{\mu_h}$ and $\kappa_2 = \frac{\Lambda_v}{\mu_v}$. It is comprehensible that F and G are locally Lipschitz concerning the initial argument and continuing in the second demonstration in $\mathbb{R}^3 \times \mathbb{R}$ and $\mathbb{R}^2 \times \mathbb{R}$, respectively. Thus, based on the existence and uniqueness theory for nonlinear ODEs, it implies that solutions of $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ exist in $(0, T)$ for some T. We can consider T as the supremum of all those types of time intervals, and if $T < \infty$, then $\lim_{t \to T^-} \|\mathbf{X}(t)\| \to \infty$ and $\lim_{t \to T^-} \|\mathbf{Y}(t)\| \to \infty$. Now, we will convey that, if

$$\mathbf{X}_0 \in \Phi_1 = \{\mathbf{X} \in \mathbb{R}^3 : X_i \geq 0 \quad \text{for} \quad i = 1, 2, 3, \sum_{i=1}^{3} X_i \leq \kappa_1\}$$

$$\mathbf{Y}_0 \in \Phi_2 = \{\mathbf{Y} \in \mathbb{R}^2 : Y_i \geq 0 \quad \text{for} \quad i = 1, 2, \sum_{i=1}^{2} Y_i \leq \kappa_2\},$$

then the sets Φ_1 and Φ_2 are positively invariant under the flow $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ for $t \in (0, T)$, respectively, and accordingly $T = \infty$, indicating that the solution persists globally in time.

Theorem 1. *The closed region*

$$\Phi_1 = \{\mathbf{X} \in \mathbb{R}^3 : X_i \geq 0 \quad \textit{for} \quad i = 1, 2, 3, \sum_{i=1}^{3} X_i \leq \kappa_1\}$$

and

$$\Phi_2 = \{\mathbf{Y} \in \mathbb{R}^2 : Y_i \geq 0 \quad \textit{for} \quad i = 1, 2, \sum_{i=1}^{2} Y_i \leq \kappa_2\}$$

are positively invariant under the flow originated by the system (2.1) and (2.2). Therefore, for initial conditions $\mathbf{X} \in \Phi_1$ *and* $\mathbf{Y} \in \Phi_2$*, solution* $\mathbf{X}(t) \in \Phi_1$ *and* $\mathbf{Y}(t) \in \Phi_2$ *remains globally in time.*

Proof. We represent boundary segments Γ_j^1, $j = 1, 2, 3, 4$ and Γ_i^2, $i = 1, 2, 3$

$$\Gamma_j^1 = \{\mathbf{X} \in \Phi_1 : X_j = 0\} \quad \text{for} \quad j = 1, 2, 3$$

$$\Gamma_4^1 = \{\mathbf{X} \in \Phi_1 : \sum_{j=1}^{4} X_j < \kappa_1\}$$

and

$$\Gamma_i^2 = \{\mathbf{Y} \in \Phi_2 : Y_i = 0\} \quad \text{for} \quad i = 1, 2$$

$$\Gamma_3^2 = \{\mathbf{Y} \in \Phi_2 : \sum_{i=1}^{3} Y_i < \kappa_2\}$$

It is clear that, $\partial\Phi_1 = \cup_{j=1}^{4} \Gamma_j^1$ and $\partial\Phi_2 = \cup_{i=1}^{3} \Gamma_i^2$.

To demonstrate the invariance of the sets Φ_1 and Φ_2, we will proclaim that, for any inward normal, $\mathbf{X}'_j.\mathbf{n}_1 \geq 0$ on $\partial\Phi_1$ and $\mathbf{Y}'_i.\mathbf{n}_2 \geq 0$ on $\partial\Phi_2$, respectively. It is apparent that the inward normal on Γ_j^1 for $j = 1, 2, 3$ and Γ_i^2 for $i = 1, 2$ are given by $\mathbf{n}_1^j = \mathbf{e}_1^j = [0, \dots, 1, \dots, 0]$ and $\mathbf{n}_2^i = \mathbf{e}_2^i = [0, \dots, 1, \dots, 0]$, where only the j & i-component is nonzero, while an inward normal on Γ_4^1 and Γ_3^2 is clearly stated by $\mathbf{n}_1^4 = (-1, -1, \dots, -1)$ and $\mathbf{n}_2^3 = (-1, -1, \dots, -1)$.

We observe that, on Γ_j^1 for $j = 1, 2, 3$,

$$\mathbf{e}_1^1 \cdot \mathbf{X}' = \Lambda_h \geq 0 \text{ for } \mathbf{X} \in \Gamma_1^1$$
$$\mathbf{e}_1^2 \cdot \mathbf{X}' = \beta_{hv} X_1 Y_2 \geq 0 \text{ for } \mathbf{X} \in \Gamma_2^1$$
$$\mathbf{e}_1^3 \cdot \mathbf{X}' = \gamma_h X_3 \geq 0 \text{ for } \mathbf{X} \in \Gamma_3^1$$

Considering, $\sum_{j=1}^{3} X_j = \kappa_1$ it satisfies,

$$\frac{dN_h}{dt} = \Lambda_h - \mu_h N_h(t),$$

Subsequently on Γ_4^1, where $N_h = \kappa_1$

$$\mathbf{n}_1^4 \cdot \mathbf{X}' = -\Lambda_h + \mu_h \kappa_1 \geq 0$$

for fixed κ_1. Therefore, Φ_1 is a positively invariant domain.

Similarly, on Γ_i^2 for $i = 1, 2$,

$$\mathbf{e}_2^1 \cdot \mathbf{Y}' = \Lambda_v \geq 0 \text{ for } \mathbf{Y} \in \Gamma_1^2$$
$$\mathbf{e}_2^2 \cdot \mathbf{Y}' = \beta_{vh} X_2 Y_1 \geq 0 \text{ for } \mathbf{Y} \in \Gamma_2^2$$

Considering, $\sum_{i=1}^{3} Y_i = \kappa_2$ it satisfies,

$$\frac{dN_v}{dt} = \Lambda_v - \mu_v N_v(t).$$

Subsequently on Γ_3^2, where $N_v = \kappa_2$

$$\mathbf{n}_2^3 \cdot \mathbf{Y}' = -\Lambda_v + \mu_v \kappa_2 \geq 0$$

for fixed κ_2. Therefore, Φ_2 is a domain with positive invariance [20].

Thus, previous remarks indicate that for initial circumstances $\mathbf{X}_0 \in \Phi_1$ and $\mathbf{Y}_0 \in \Phi_2$, solution $\mathbf{X}(t) \in \Phi_1$ and $\mathbf{Y}(t) \in \Phi_2$ exist globally in time. □

Remarks. The steady-state points and basic reproduction number regarding the model have been calculated briefly in Appendix A. After simplification, the required DFE (disease-free equilibrium) point, basic reproduction number ($\mathfrak{R}_{00}$) and endemic equilibrium (E^*) of the corresponding model become:

$$\text{DFE, } P_0 = \left(\frac{\Lambda_h}{\mu_h}, 0, \frac{\Lambda_v}{\mu_v}, 0\right),$$

$$\mathfrak{R}_{00} = \left(\frac{\beta_{hv}\dfrac{N_v}{N_h}}{\gamma_h + \mu_h}\right)\left(\frac{\beta_{vh}}{\mu_v}\right),$$

and

$$E^* = (S_h^*, I_h^*, S_v^*, I_v^*)$$

where,

$$S_h^* = \frac{\Lambda_h N_h Q_1}{\mu_h N_h Q_1 + \beta_{hv} Q_2 I_h^*}, \quad S_v^* = \frac{N_h \Lambda_v}{\mu_v N_h + \beta_{vh} I_h^*},$$

$$I_v^* = \frac{Q_2 I_h^*}{N_h \mu_v^2 + \mu_v \beta_{vh} I_h^*}, \quad \text{and}$$

$$f(I_h^*) = P_1 I_h^{*2} + P_2 I_h^* + P_3 = 0.$$

The abbreviation of $Q_1, Q_2, Q_3, Q_4, Q_5, P_1, P_2, P_3$ are presented in the appendix. Furthermore, stability analysis of the steady-state points is considered in Appendix B.

Table 2
The LHS analysis used baseline, maximum, and minimum values.

Parameter	Minimum	Baseline	Maximum
Λ_h	0.00001	0.000039	0.005
Λ_v	0.009	0.143	0.4
μ_h	0.0001	0.00039	0.009
μ_v	0.088	0.1	0.13
γ_h	0.15	0.2	0.302
β_{hv}	0.19	0.225	0.28
β_{vh}	0.312	0.375	0.415

3. Monte Carlo sampling technique (sensitivity analysis)

The LHS/PRCC procedure is applied to numerous mathematical models with continuous or discrete properties that are deterministic or stochastic. The Partial Rank Correlation analysis provides us with a PRCC and proportional p-values which help to determine the amount of uncertainty that an LHS parameter lends to the prototype [35–37]. The magnitude along with the statistical importance of the parameter's PRCC value recommends that the parameter's contribution to forecasting the model is imprecision. The most significant parameters have large PRCC values (>0.5 or < − 0.5) and correspondingly small p-values (<0.05). The LHS parameter has a stronger effect on the outcome measure the nearer the PRCC value is to +1 or −1. The sign denotes how qualitatively different the input and output variables are from one another. The LHS parameter is inversely related to the result measure if it has a negative sign.

Maximum and minimum values are dictated for each of the seven LHS parameters in Table 2 which are reliant on our considered model. It should be noted that the baseline value for each LHS parameter has been assigned to a value that is in the midpoint of the range between the parameter's minimum and maximum values.

Some of the baseline value of the parameter is estimated, and some of them are taken from the reference, which is presented in Table 6. In Table 2, each parameter's minimum and maximum values are estimated based on best or well-fitted curves.

In this section, we carried out a sensitivity analysis to determine the model (2.1) and (2.2) robustness to parameter values, which helps to identify the most significant parameters in model dynamics. A Monte Carlo Sampling technique is used, which is widely known as the Latin Hypercube Sampling (LHS) scheme that samples 1000 values for each input parameter using a uniform distribution over the range of biologically realistic values described in Table 2. By randomly selecting pairing sampled values for all LHS parameters, 5000 model simulations were performed for the system of differential equations described in (2.1) and (2.2). Fig. 3 shows the corresponding nonlinear but unmodulated relation between model state variables and each parameter by the Partial Rank Correlation Coefficients (PRCC) and corresponding p-values. Moreover, the output is statistically significant if the corresponding p-value is less than 1%.

From Fig. 3, we can identify parameters that strongly influence the model dynamics, namely μ_v, γ_h, and β_{vh}. All significant parameters have a mixed influence on the model variables. We accomplish a multilinear regression analysis of the ranks achieved for the consequence measures (i.e., total infected hosts and infected vectors) and the LHS parameters. Then, we undertake a regression analysis on these ranks to examine the regression coefficients. We move to establish the strength of the association between the LHS parameter and each consequence measure by collecting the PRCC values since these regression coefficients show how sensitive the model is to the LHS parameters. Figs. 4 and 5 show PRCC plots, including PRCC value and p-value for total infected hosts and vectors. The residuals for the ranked LHS parameter values are plotted on the y-axis in Figs. 4 and 5, respectively. We have also plotted the residuals for the ranked total infected hosts on the x-axis of Fig. 4, and the residuals for the total infected vectors on the x-axis of Fig. 5. Additionally, notice how various LHS factors exhibit a high association, as shown by the PRCC graphs for the two consequence measures.

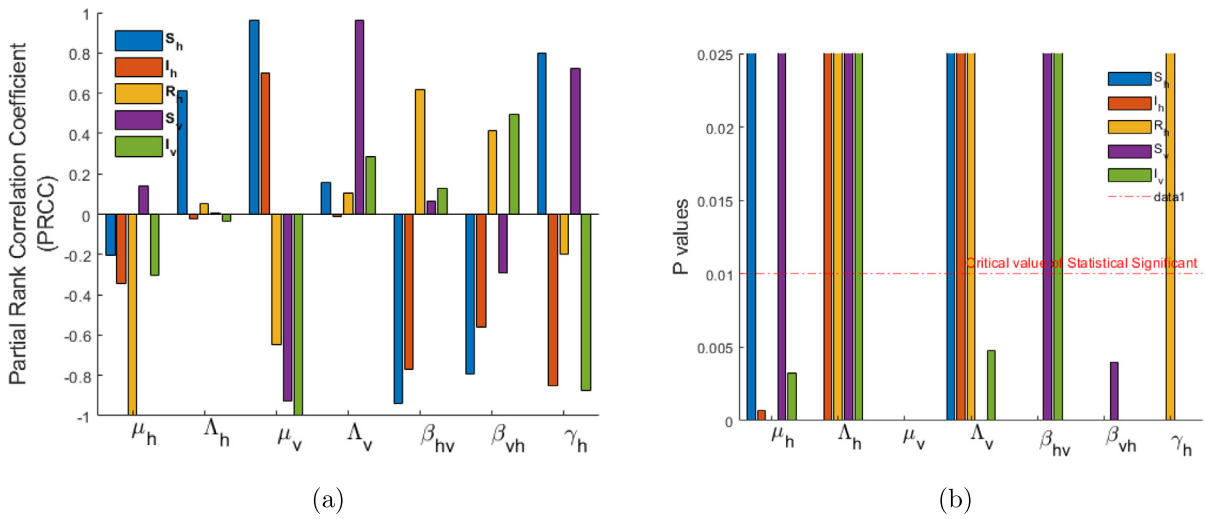


Fig. 3. PRCC and P-values plot for the vector-host model.

Table 3
Output from PRCC analysis for host.

Parameter	Susceptible host		Infected host		Recovered host	
	PRCC	p-value	PRCC	p-value	PRCC	p-value
Λ_h	0.44353	3.8065e−06	0.25396	0.010785	0.023579	0.81587
Λ_v	0.0624	0.5374	−0.16531	0.10024	0.075184	0.45722
μ_h	−0.12408	0.21872	−0.39074	5.839e−05	***−0.99742	5.887e−114
μ_v	***0.92575	3.5283e−43	**0.70123	4.5041e−16	−0.42229	1.2069e−05
γ_h	*0.67198	1.9497e−14	***−0.88692	1.214e−34	−0.25361	0.010898
β_{hv}	***−0.85809	4.0146e−30	***−0.81228	1.1187e−24	0.46411	1.1532e−06
β_{vh}	*−0.65206	2.0063e−13	*−0.60134	3.6992e−11	0.34689	0.0004066

Table 4
PRCC outcomes for vector population.

Parameter	Susceptible vector		Infected vector	
	PRCC	p-value	PRCC	p-value
Λ_h	−0.12688	0.20841	−0.19286	0.05455
Λ_v	***0.9568	2.2188e−54	0.25991	0.0090147
μ_h	0.077677	0.44239	−0.35551	0.00028373
μ_v	***−0.92532	4.6398e−43	***−0.99849	2.2932e−125
γ_h	**0.7535	1.4978e−19	***−0.92047	8.9894e−42
β_{hv}	0.14885	0.13939	−0.18869	0.060094
β_{vh}	−0.48269	3.6598e−07	**0.7056	2.4681e−16

3.1. Analyzing the PRCC results

The outcomes from the PRCC diagrams are encapsulated in Tables 3 and 4. In those tables, the PRCC values (blue) and p-values (orange) of significant uncertainty factors are emphasized, not just one or the other. Here, (*) is utilized to denote possible contributors (PRCC values: 0.5 to 0.69 or −0.5 to −0.69), (**) has been used to identify the very likely sources of ambiguity (PRCC values: 0.7 to 0.79 or −0.7 to −0.79), and (***) is used to recognize the extremely likely origins of uncertainty (PRCC values: 0.8 to 0.99 or −0.8 to −0.99).

From Tables 3 and 4, we remark that the most significant LHS parameters for the consequence measure are the following:

- μ_v, death rate of the vector.
- γ_h, host recovery rate.
- β_{vh}, transmission rate from vector to host.

From Tables 3 and 4, it is also noticeable that the PRCC values of μ_v are higher than those of the other parameters, and conclude that μ_v is the most prominent parameters in the considered model.

4. Formulation of continuous time Markov chain (CTMC) model

Malaria dynamics are essentially stochastic due to multiple factors, including mosquito-biting behavior, parasite acquisition, human immune responses, and the efficacy of therapies. The continuous-time Markov chain (CTMC) model incorporates random events, enabling it possible to simulate these stochastic processes and capture the variability apparent in realistic malaria dynamics. Malaria infection progression involves discrete states such as susceptible, infected, and recovered. CTMC models are convenient for depicting such discrete states and the transition between them, making them acceptable for observing the dynamics of malaria infection over time. Furthermore, the CTMC model is capable of including individual-level heterogeneity, enabling the simulation of host populations and capturing the effect of heterogeneity on disease transmission and consequences.

CTMC models utilize transition rates between states, and these rates determine the probability of transitioning from one state to another regarding the specific biological processes implicated in malaria dynamics. On the other hand, the CTMC model can represent time-dependent events, including infected mosquito arrival, bite incidents, parasite growth, and recovery procedures. This makes it possible to describe disease states' stochastic and continuous evolution throughout time.

Hence, the CTMC models are suitable for malaria research because they capture the disease's complexity, stochasticity, and temporal dynamics. CTMC model makes it easier to simulate discrete infection states, time-dependent events, and individual-level variation, thereby providing a more realistic representation of malaria transmission and infection dynamics.

The Markov property implies that the translation of any state variable to the state i_{n+1} at time t_{n+1} depends on the value of the current state at time t_n rather than the filtration or history of the process. In the CTMC model, the process is time-homogeneous with Markov property

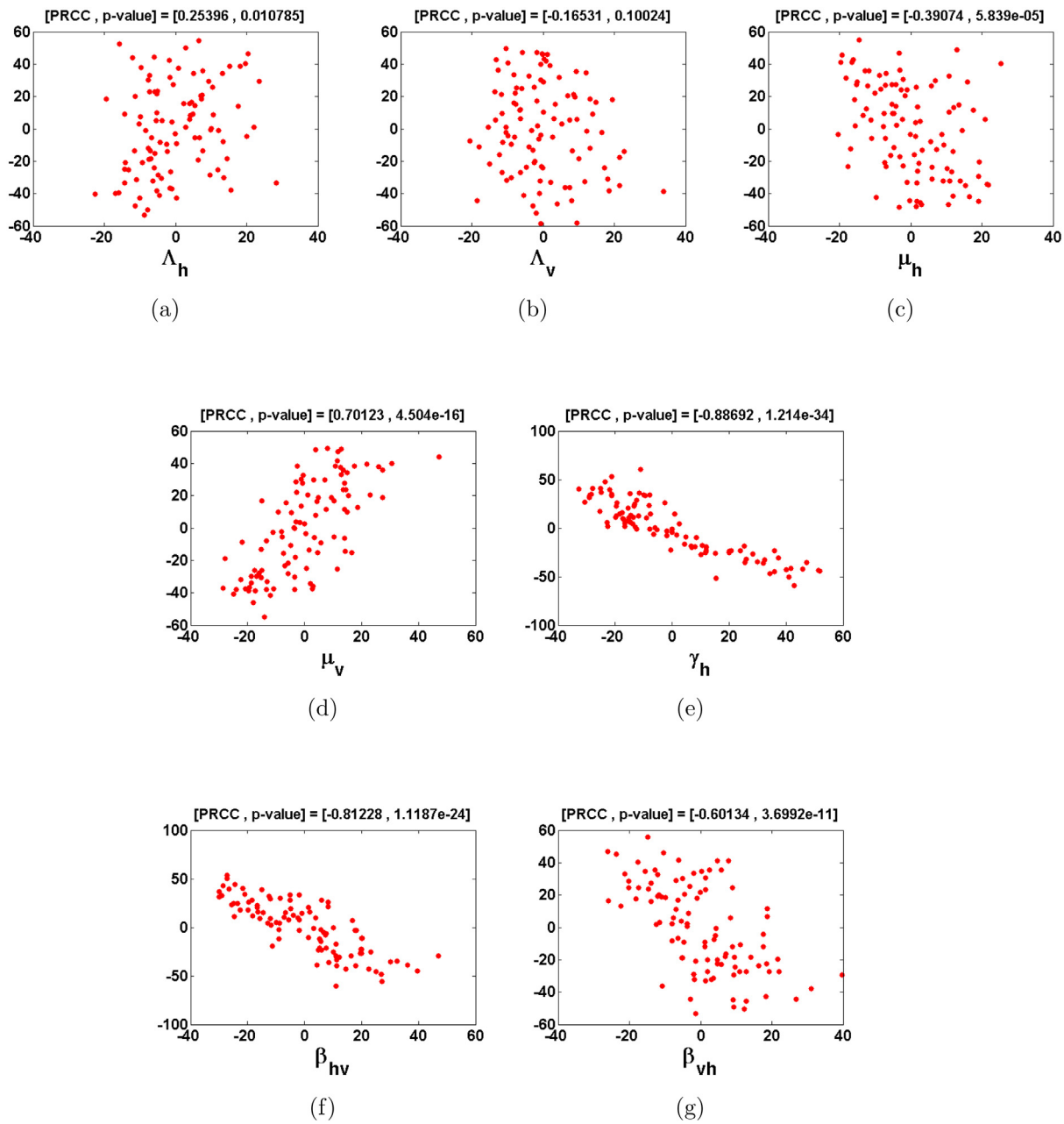


Fig. 4. PRCC plots for infected hosts.

and the state variables are discrete-valued where time is assumed as continuous.

We will convert the deterministic ODE system for malaria to the CTMC model. For our convenience and understanding, we will use the same notation as the deterministic ODE model for both state variables and model parameters in CTMC model [16,38].

Let, $\mathfrak{Q}(\mathrm{t}) = \left(S_h(t), I_h(t), R_h(t), S_v(t), I_v(t)\right)$. For the CTMC model, the random variables, $S_h(t), I_h(t), R_h(t), S_v(t), I_v(t)$ are discrete and time is continuous, $t \in [0, \infty)$. So,

$$S_h(t), I_h(t), R_h(t), S_v(t), I_v(t) \quad \in \quad \{0, 1, 2, 3, 4, \ldots\}$$

Suppose, there exists a sufficiently small time interval $\Delta t > 0$ such that at most one incident happens during the time interval Δt. Let,

$$\Delta\mathfrak{Q}(\mathrm{t}) = \left(\Delta S_h(t), \Delta I_h(t), \Delta R_h(t), \Delta S_v(t), \Delta I_v(t)\right),$$

where, $\Delta S_h = (S_h(t+\Delta t) - S_h(t))$, $\Delta I_v = (I_v(t+\Delta t) - I_v(t))$ etc. That means, it denotes the change of the system variables over a small time interval Δt. Depend on the compartment system (2.1) and (2.2), we summarize various events along with their corresponding transition probabilities in Table 5. The first event of Table 5 represents the birth of a healthy host, event 5 is the infection of the host, etc. Specifically, the probability of event 9 equals,

$$\mathbb{P}(\Delta\mathfrak{Q}(t) = (0, 0, 0, -1, 1)|\mathfrak{Q}(\mathrm{t})) = \beta_{vh} S_v \frac{I_h}{N_h} \Delta t + o(\Delta t).$$

Implementing the CTMC model for malaria research can present technical challenges. To develop a detailed CTMC model for malaria, we have to consider various parameters, events, and state transitions which can make the model intricate and potentially computationally intensive. Moreover, running simulations of the CTMC model

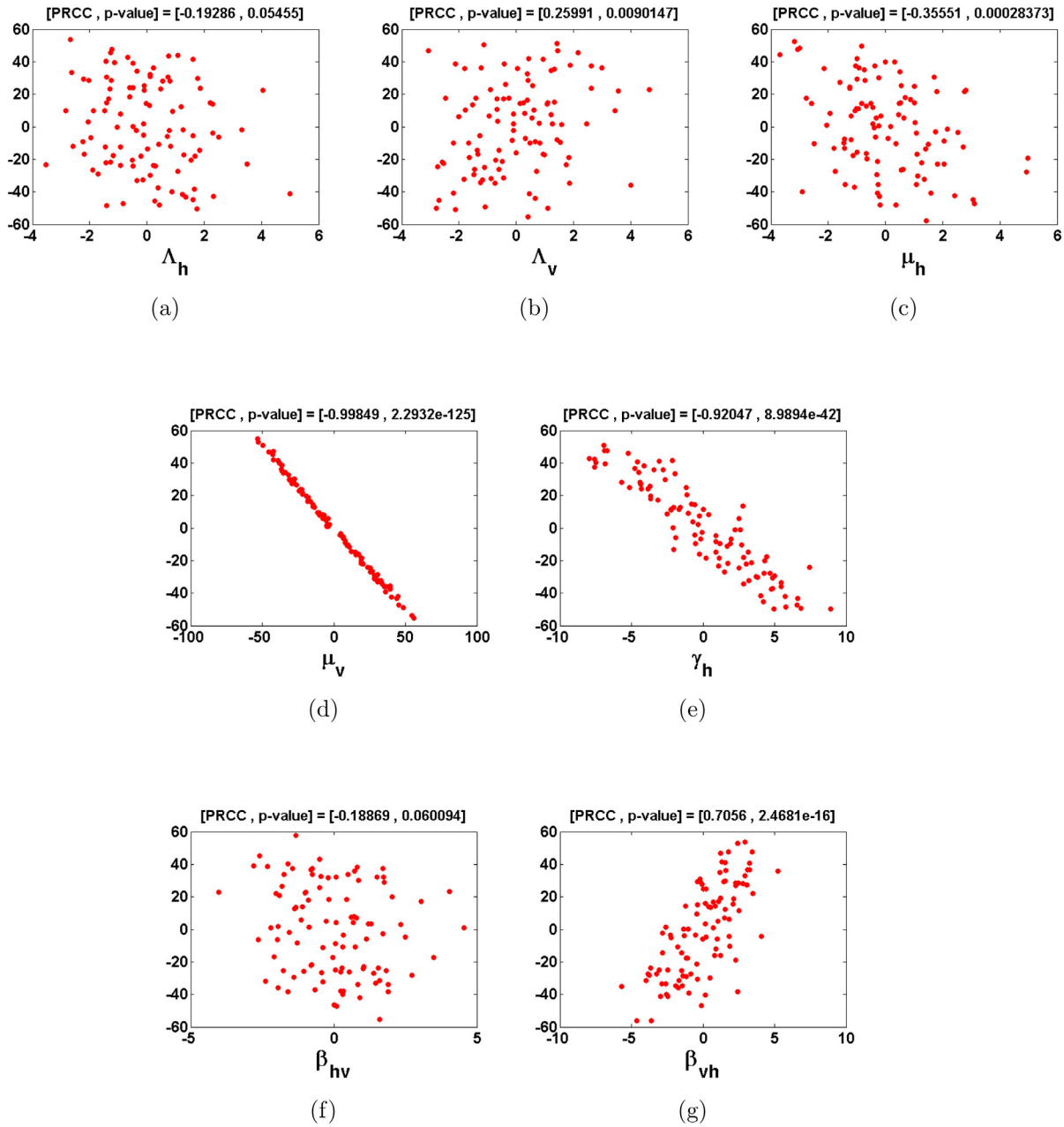


Fig. 5. PRCC plots for infected vectors.

Table 5
Possible changes of the state variables during the time interval Δt with corresponding transition probabilities for the CTMC model.

Event, i	Description	$\Delta\mathfrak{Q}(t)$	Transition probabilities, $p_i\Delta t + o(\Delta t)$
1	Healthy host birth	$(1,0,0,0,0)$	$\Lambda_h\Delta t + o(\Delta t)$
2	Healthy host death	$(-1,0,0,0,0)$	$S_h\mu_h\Delta t + o(\Delta t)$
3	Healthy vector birth	$(0,0,0,1,0)$	$\Lambda_v\Delta t + o(\Delta t)$
4	Healthy vector death	$(0,0,0,-1,0)$	$\mu_v S_v\Delta t + o(\Delta t)$
5	Infection of host	$(-1,1,0,0,0)$	$\beta_{hv} S_h \frac{I_v}{N_h}\Delta t + o(\Delta t)$
6	Death of infected host	$(0,-1,0,0,0)$	$\mu_h I_h\Delta t + o(\Delta t)$
7	Infected host recovery	$(0,-1,1,0,0)$	$\gamma_h I_h\Delta t + o(\Delta t)$
8	Recovered host death	$(0,0,-1,0,0)$	$\mu_h R_h\Delta t + o(\Delta t)$
9	Infection of vector	$(0,0,0,-1,1)$	$\beta_{vh} S_v \frac{I_h}{N_h}\Delta t + o(\Delta t)$
10	Infected vector death	$(0,0,0,0,-1)$	$\mu_v I_v\Delta t + o(\Delta t)$
11	No change	$(0,0,0,0,0)$	$1-\sum_{i=1}^{10} p_i\Delta t + o(\Delta t)$

for extended periods or large-scale populations may require substantial computational resources and efficient algorithms to handle the computational burden.

5. Formulation of SDE model

Now, we will derive an Itô stochastic differential equations (SDEs) according to the CTMC model. Here, we will use the same notation for each parameter and variable as in the deterministic system and CTMC model. In SDE model, both random variables and time are continuous and lie in the set $[0, \infty)$ [16,38,39]. Within a short period Δt, the Itô SDEs for the model is formulated by computing the expected changes $u(t)$ and covariance $C(t)$ of $\Delta\mathfrak{Q}$ depended on the transition probabilities of each event in Table 5. The system of SDEs leads as follows:

$$d\mathfrak{Q} = u(t) + \mathcal{G}(t)dW(t)$$

where, $u(t)$ represents the drift vectors, $\mathcal{G}(t)$ is the diffusion matrix which satisfies $\mathcal{G}(t)\mathcal{G}^T(t) = C(t)$, and $W(t)$ is a vector of 10 independent Brownian or Wiener processes. In particular, let $\mathcal{G} = (\breve{\mathcal{A}}|\breve{\mathcal{B}})$ where $\breve{\mathcal{A}}$ and $\breve{\mathcal{B}}$ both are 5×5 matrix, respectively as follows:

$$\breve{\mathcal{A}} = \begin{pmatrix} \sqrt{\mu_h N_h} & -\sqrt{S_h \mu_h} & 0 & 0 & -\sqrt{\beta_{hv} S_h \frac{I_v}{N_h}} \\ 0 & 0 & 0 & 0 & \sqrt{\beta_{hv} S_h \frac{I_v}{N_h}} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{\mu_v N_v} & -\sqrt{\mu_v S_v} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{5.1}$$

$$\breve{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ -\sqrt{\mu_h I_h} & -\sqrt{\gamma_h I_h} & 0 & 0 & 0 \\ 0 & \sqrt{\gamma_h I_h} & -\sqrt{\mu_h R_h} & 0 & 0 \\ 0 & 0 & 0 & -\sqrt{\beta_{vh} \frac{I_h}{N_h} S_v} & 0 \\ 0 & 0 & 0 & \sqrt{\beta_{vh} \frac{I_h}{N_h} S_v} & -\sqrt{\mu_v I_v} \end{pmatrix} \tag{5.2}$$

Thus more specifically, the system of SDEs for vector-host malaria model is,

$$\begin{aligned} dS_h(t) &= u_1 dt + \sqrt{\mu_h N_h} W_1 - \sqrt{S_h \mu_h} W_2 - \sqrt{\beta_{hv} S_h \frac{I_v}{N_h}} W_5 \\ dI_h(t) &= u_2 dt + \sqrt{\beta_{hv} S_h \frac{I_v}{N_h}} W_5 - \sqrt{\mu_h I_h} W_6 - \sqrt{\gamma_h I_h} W_7 \\ dR_h(t) &= u_3 dt + \sqrt{\gamma_h I_h} W_7 - \sqrt{\mu_h R_h} W_8 \\ dS_v(t) &= u_4 dt + \sqrt{\mu_v N_v} W_3 - \sqrt{\beta_v h \frac{I_h}{N_h} S_v} W_9 \\ dI_v(t) &= u_5 dt + \sqrt{\beta_v h \frac{I_h}{N_h} S_v} W_9 - \sqrt{\mu_v I_v} W_{10} \end{aligned} \tag{5.3}$$

where, the column vector $u = (u_1, u_2, u_3, u_4, u_5)^T$ comes directly from deterministic ODE model, that means:

$$u = \begin{pmatrix} \Lambda_h - \beta_{hv} I_v \frac{S_h}{N_h} - \mu_h S_h \\ \beta_{hv} I_v \frac{S_h}{N_h} - (\gamma_h + \mu_h) I_h \\ \gamma_h I_h - \mu_h R_h \\ \Lambda_v - \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v S_v \\ \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v I_v \end{pmatrix} \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{matrix} \tag{5.4}$$

6. Result and discussion

The appropriate parameter values for the desired vector host model (2.1) and (2.2) are included in Table 6. The rate of disease transmission from host to vector and vector to host is important in controlling malaria outbreaks.

Figs. 6(a), 6(b), and 6(c) depict the solution of the ODE model (2.1) and (2.2) with parameter values from the Table 6 and the initial conditions $S_h(0) = 99$, $I_h(0) = 1$, $R_h(0) = 0$, $S_v(0) = 999$, $I_v(0) = 1$, and take into account the total number of hosts, and vectors are $N_h = 100$, and $N_v = 1000$. In particular, deterministic solutions for host and vector populations have been shown in the diagrams 6(a) and 6(b), respectively. Based on those illustrations, we can conclude that the recovered subjective increases (the green line) while the infected population for the host as well as vector decreases over time. Furthermore, we are concerned about the behavior of the infected individuals. In Fig. 6(c), the infected host (blue smooth curve) and vector (red smooth curve) are drawn, and it shows that after a certain period, the infected host and vector curve hit the x-axis. Thus, the infected host and vector vanish gradually.

For capturing variability and inherent randomness in biological systems, stochastic models play a vital role in biological research. This is especially important when researching infectious diseases, ecological systems, or population dynamics involving high individual variation, chance occurrences, and random interactions. One of the key benefits of employing a stochastic model over a deterministic model is its capacity to consider the inherent randomness and uncertainties in biological processes. A deterministic model ignores stochastic effects and presumes known initial circumstances and deterministic relationships between variables. The stochastic model provides a more accurate depiction of the dynamics since the model can take into account chance occurrences, random fluctuations, and individual-level heterogeneity that can significantly influence the results. This research introduces a novel perspective and contributes to the field by providing a more comprehensive and realistic analysis of the malaria model under investigation by integrating stochasticity.

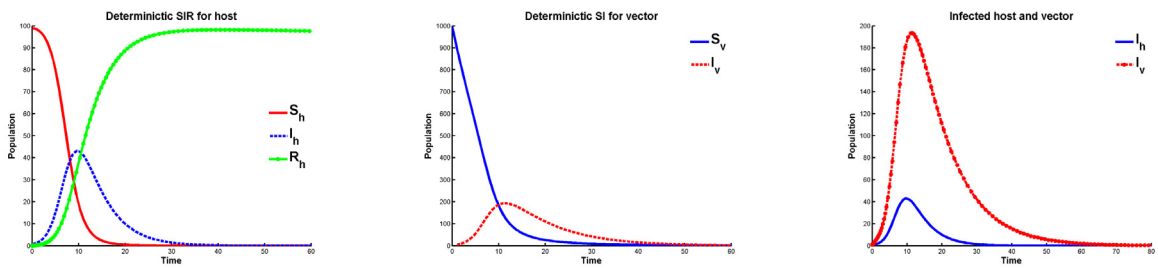
We would like to see how the solution of the ODE models (2.1) and (2.2) behaves stochastically. For the CTMC model, Table 5 presents transition probabilities of each event corresponding to the model (2.1) and (2.2). Figs. 7(a), 7(b), 8(a) and 8(b) show the dynamics of host and vector populations for the CTMC model. They represent three sample paths for host and vector individuals. Moreover, a closer view of the infected host and infected vector population is exposed in Figs. 7(c), 7(d), 8(c) and 8(d). Based on diagram 7(a), it is predictable that the net peak of the infected host occurs at 30 to 40 days, with the total number of host and infected individuals at 100 and 10, respectively. The outcomes in Fig. 7(b) display that the number of infected vectors reduces rapidly within 60 to 80 days corresponding to the initial number of infected vectors at 60.

Additionally, by increasing the number of host and vector populations, the number of infected host gain meridian position within 35 to 50 days, Fig. 8(a). Besides, the number of infected vectors decrease faster within 80 to 100 days, Fig. 8(b).

System (5.3) is the corresponding SDE model of (2.1) and (2.2). In the schematics 9(a) and 9(b), we draw the system (5.3) for the host and vector. By adopting a modest time interval in which at least one event occurs, Fig. 9(a) shows three sample paths of each host compartment. We utilize the same parameter values and several participants for the graphs 9(a) and 9(b). Sample paths for the infected host (red curve) show that infection will fade for the time being, and the number of recovered hosts (black curve) will be nearly equal to the total number of individuals we have collected. In addition, Fig. 9(b) shows three sample pathways for susceptible and infected vectors. With time, the number of affected individuals will drop significantly. We also know that the curve of CTMC and SDE is continuous but not differentiable. Moreover, SDE converges to its approximate solution faster than the CTMC model, providing a better approximation. It is obvious that disease transmission takes longer in larger populations. In this work, we observe the model output for the deterministic and CTMC models and examine the outcome of the model's corresponding stochastic differential equation.

Table 6
Model parameter values.

Parameter	Clarification	Units	Value	Source
Λ_h	Host birth rate	Per day	0.000039	[4]
Λ_v	Vector birth rate	Per day	0.143	[4]
μ_h	Host death rate	Per day	0.00039	[4]
μ_v	Vector death rate	Per day	0.1	[4]
γ_h	Recovery rate of host	Per day	0.2	Estimated
β_{hv}	Transmission probability from an infectious vector	Per day	0.225	Estimated
β_{vh}	Transmission probability from an infectious host	Per day	0.375	Estimated

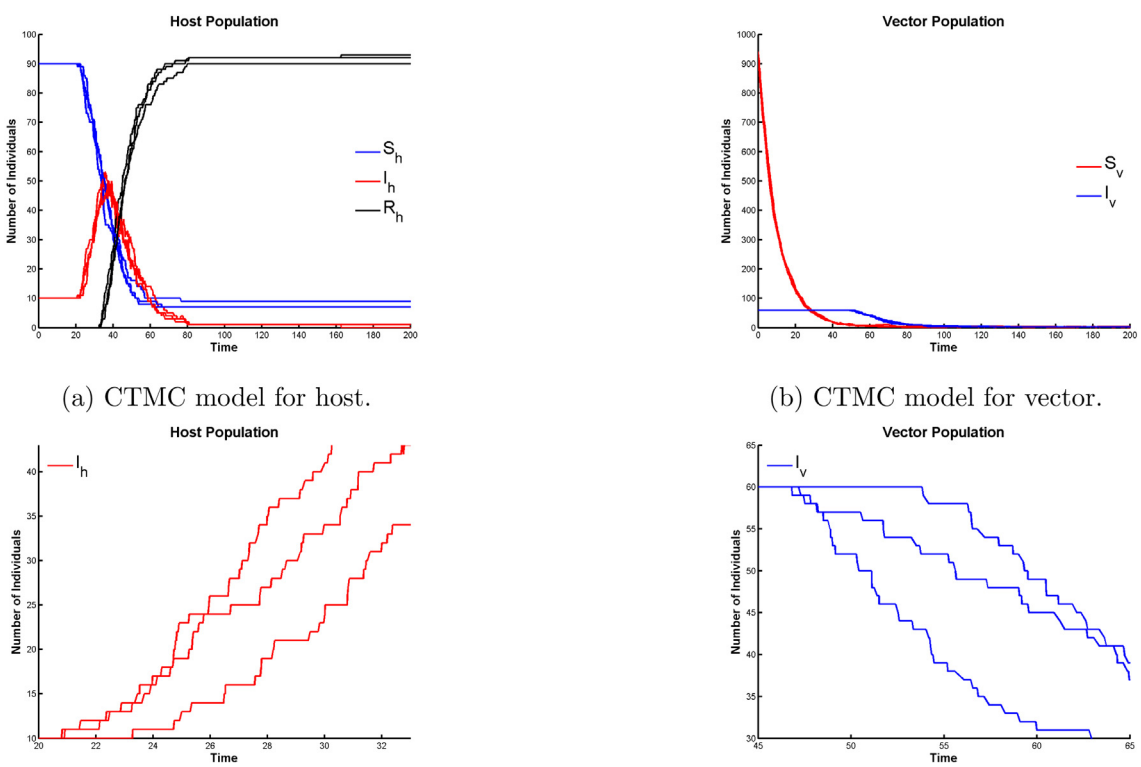


(a) Deterministic behaviour of host population.

(b) Deterministic behaviour of vector population.

(c) Deterministic behaviour of infected host and vector.

Fig. 6. Dynamics of the vector-host model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) CTMC model for host.

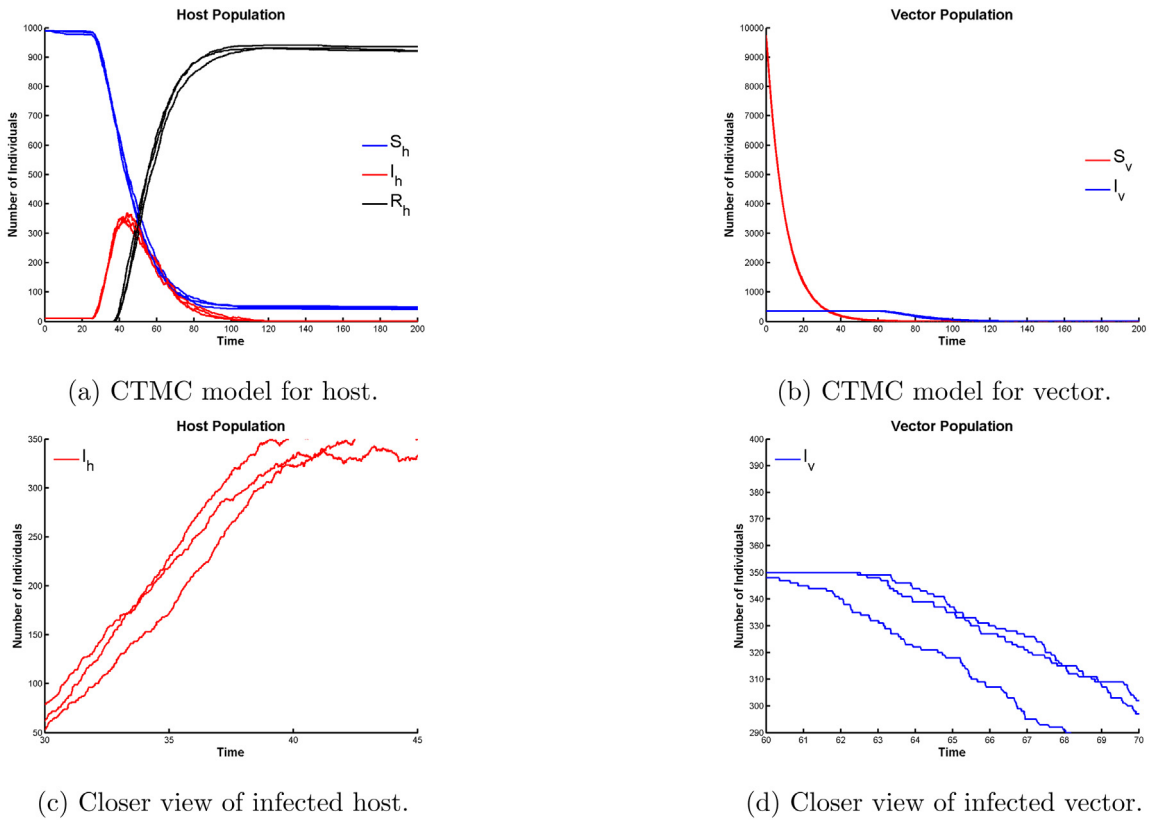
(b) CTMC model for vector.

(c) Closer view of infected host.

(d) Closer view of infected vector.

Fig. 7. The curves represent three sample paths in the vector-host CTMC model. The parameter values are $\Lambda_h = 0.000039$, $\Lambda_v = 0.143$, $\mu_h = 0.00039$, $\mu_v = 0.1$, $\gamma_h = 0.2$, $\beta_{hv} = 0.225$, $\beta_{vh} = 0.375$ and $N_h = 100$. The starting conditions are $S_h(0) = 90$, $I_h(0) = 10$, $R_h(0) = 0$, $S_v(0) = 940$ and $I_v(0) = 60$.

model based on human SI insect ☆ can make model



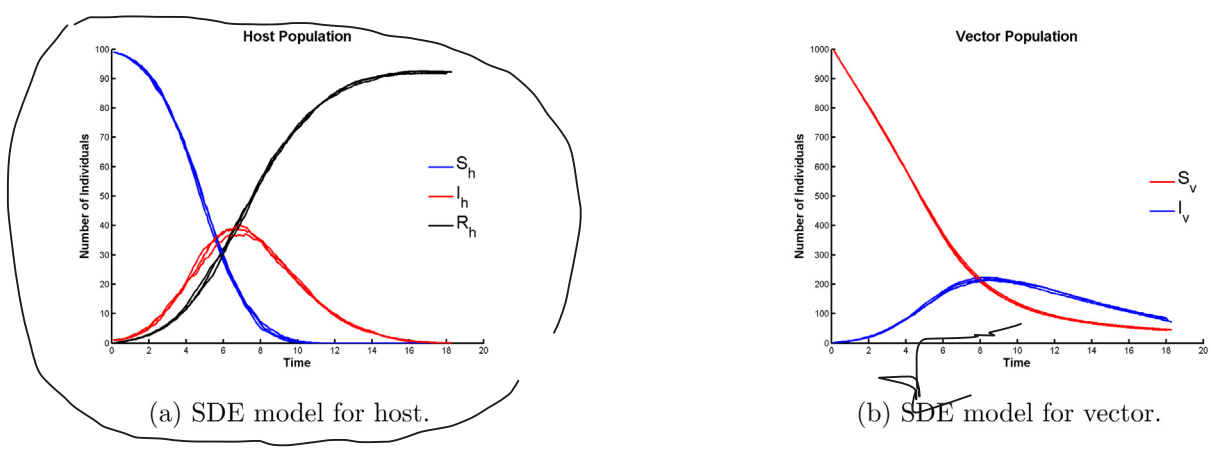
(a) CTMC model for host.

(b) CTMC model for vector.

(c) Closer view of infected host.

(d) Closer view of infected vector.

Fig. 8. The curves represent three sample paths in the vector-host CTMC model. The parameter values are $\Lambda_h = 0.000039$, $\Lambda_v = 0.143$, $\mu_h = 0.00039$, $\mu_v = 0.1$, $\gamma_h = 0.2$, $\beta_{hv} = 0.225$, $\beta_{vh} = 0.375$ and $N_h = 1000$. The starting conditions are $S_h(0) = 990$, $I_h(0) = 10$, $R_h(0) = 0$, $S_v(0) = 9750$ and $I_v(0) = 350$.



(a) SDE model for host.

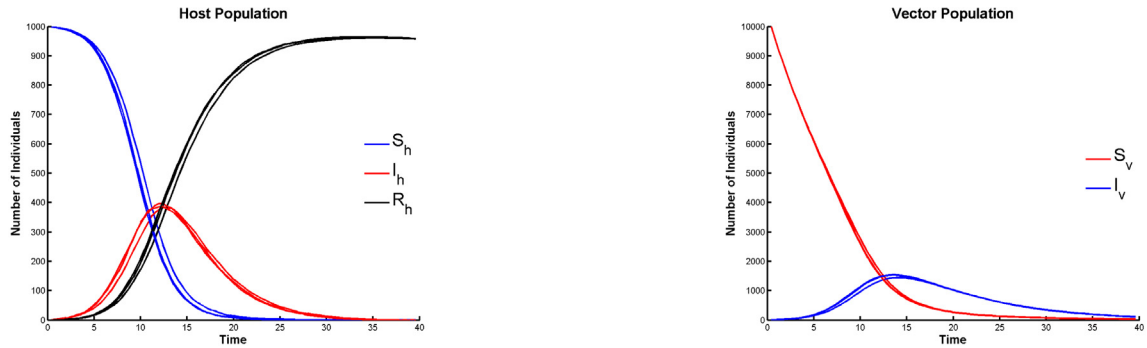
(b) SDE model for vector.

Fig. 9. The curves represent three sample paths in the vector-host SDE model. The parameter values are $\Lambda_h = 0.000039$, $\Lambda_v = 0.143$, $\mu_h = 0.00039$, $\mu_v = 0.1$, $\gamma_h = 0.2$, $\beta_{hv} = 0.225$, $\beta_{vh} = 0.375$ and $N_h = 100$. The starting conditions are $S_h(0) = 99$, $I_h(0) = 1$, $R_h(0) = 0$, $S_v(0) = 999$ and $I_v(0) = 1$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

All the observations provide us with a similar conclusion. Thus, it ensures that the CTMC model accurately represents malaria dynamics and produces reliable and meaningful results, supporting its application in malaria research (see Fig. 10).

7. Concluding remarks

We now live in a society where multidisciplinary research is the most critical mechanism for humanity to live a healthy and standard existence. Over the years, mathematical prediction modeling has been one of the essential techniques for studying and understanding the behavior of epidemics. It enables policymakers to make critical decisions and plan for the next few days. In this study, we consider the malaria model where the SIR epidemic model is used for host populations and SI dynamical system is approached for vector individuals. We have considered a sensitivity analysis for each parameter of the model. After analyzing PRCC values and p-values, we finally conclude that the natural death rate μ_v is the most sensitive parameter for controlling disease outbreaks. If we increase the natural death rate of vectors (mosquitoes), there will be less chance of spreading malaria transmission. Deterministic and stochastic dynamics are established for the malaria model after considering all important aspects of disease progression. We establish theoretical results and represent them graphically in the CTMC and SDE stochastic model. It is evident that an acceptable agreement is found between the dynamical behavior of the deterministic and stochastic models. Furthermore, it is demonstrated that the number of infected individuals will be reduced gradually after a certain period. Thus, our desired model (deterministic and stochastic) assures the strategy's



(a) SDE model for host. (b) SDE model for vector.

Fig. 10. The curves represent three sample paths in the vector-host SDE model. The parameter values are $\Lambda_h = 0.000039$, $\Lambda_v = 0.143$, $\mu_h = 0.00039$, $\mu_v = 0.1$, $\gamma_h = 0.2$, $\beta_{hv} = 0.225$, $\beta_{vh} = 0.375$ and $N_h = 1000$. The starting conditions are $S_h(0) = 999$, $I_h(0) = 1$, $R_h(0) = 0$, $S_v(0) = 9990$ and $I_v(0) = 10$.

effectiveness in reducing and gradually preventing disease outbreaks. The investigation can be expanded in the future by examining the cross-transmission of mosquitoes. Another expansion of the research could be to consider the consequences of environmental and seasonal variability on disease transmission trajectories, such as seasonal rainfall, heating rate, and moisture levels. Someone may include the emigration and immigration of adults in both groups to examine the influence of mobility on disease transmission [40].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Steady states

A.1. Disease-free equilibrium

To compute equilibrium point, setting all the time derivatives equal to zero [20], we get,

$$\begin{cases} \Lambda_h - \beta_{hv} I_v \frac{S_h}{N_h} - \mu_h S_h = 0, \\ \beta_{hv} I_v \frac{S_h}{N_h} - (\gamma_h + \mu_h) I_h = 0, \\ \Lambda_v - \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v S_v = 0, \\ \beta_{vh} S_v \frac{I_h}{N_h} - \mu_v I_v = 0. \end{cases} \tag{A.1}$$

At disease-free equilibrium point $I_h = 0$, and $I_v = 0$ since we are assuming that there is no disease at disease-free equilibrium point. Then,

$S_h = N_h$ and $S_v = N_v$

by solving first and third equation we get,

$$\Lambda_h - \mu_h S_h = 0 \Rightarrow S_h = \frac{\Lambda_h}{\mu_h} \quad \text{and} \quad \Lambda_v - \mu_v S_v = 0 \Rightarrow S_v = \frac{\Lambda_v}{\mu_v}.$$

Thus, the disease-free equilibrium point is,

$$P_0 = \left(\frac{\Lambda_h}{\mu_h}, 0, \frac{\Lambda_v}{\mu_v}, 0 \right).$$

A.2. Basic reproduction number

At the disease-free equilibrium (DFE) point, we will find the basic reproduction number of the model (2.4) by using Next generation approach as well as the Jacobian approach.

At first, we apply the next-generation approach for the compartment models to find out the basic reproduction number. In 1990, the next generation matrix was introduced by Diekmann and Heester-beek where the spectral radius of the next generation matrix is the basic reproduction number, $\mathfrak{R}_0$ [1].

For the next-generation approach in vector-host model (2.4), suppose

$$\mathcal{F}_i = \begin{pmatrix} \beta_{hv} I_v \frac{S_h}{N_h} \\ \beta_{vh} S_v \frac{I_h}{N_h} \end{pmatrix} \quad \textit{and} \quad \mathcal{V}_j = \begin{pmatrix} (\gamma_h + \mu_h) I_h \\ \mu_v I_v \end{pmatrix} \qquad \text{for} \quad i, j = 1, 2$$

Here, $\mathcal{F}_i$ is the rate of newly infected individuals appearance in the ith compartment. $\mathcal{V}_j$ is the remaining transmission terms like births, recovery, and deaths.

At DFE point, $S_h = N_h$ and $S_v = N_v$.

So, the matrix F and V is computed as the Jacobian matrix of $\mathcal{F}_i$ and $\mathcal{V}_j$, respectively at disease-free equilibrium point,

$$F = \begin{pmatrix} \frac{\partial \mathcal{F}_1}{\partial I_h} & \frac{\partial \mathcal{F}_1}{\partial I_v} \\ \frac{\partial \mathcal{F}_2}{\partial I_h} & \frac{\partial \mathcal{F}_2}{\partial I_v} \end{pmatrix} = \begin{pmatrix} 0 & \beta_{hv} \frac{S_h}{N_h} \\ \beta_{vh} \frac{S_v}{N_h} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \beta_{hv} \\ \beta_{vh} \frac{N_v}{N_h} & 0 \end{pmatrix} \quad \text{and}$$

$$V = \begin{pmatrix} \frac{\partial \mathcal{V}_1}{\partial I_h} & \frac{\partial \mathcal{V}_1}{\partial I_v} \\ \frac{\partial \mathcal{V}_2}{\partial I_h} & \frac{\partial \mathcal{V}_2}{\partial I_v} \end{pmatrix} = \begin{pmatrix} \gamma_h + \mu_h & 0 \\ 0 & \mu_v \end{pmatrix}. \quad \text{So,} \quad V^{-1} = \begin{pmatrix} \frac{1}{\gamma_h + \mu_h} & 0 \\ 0 & \frac{1}{\mu_v} \end{pmatrix}$$

$$FV^{-1} = \begin{pmatrix} 0 & \frac{\beta_{hv}}{\mu_v} \\ \frac{\beta_{vh} \frac{N_v}{N_h}}{\gamma_h + \mu_h} & 0 \end{pmatrix}$$

Thus, the spectral radius of FV^{-1} is,

$$\mathfrak{R}_0 = \sigma(FV^{-1}) = \sqrt{\left(\frac{\beta_{hv}}{\mu_v} \right) \left(\frac{\beta_{vh} \frac{N_v}{N_h}}{\gamma_h + \mu_h} \right)}$$

which is our required basic reproduction number ($\mathfrak{R}_0$) for the model (2.4).

At the disease-free equilibrium (DFE) point, the reproduction number represents a threshold condition to analyze the stability of infectious disease model.

The DFE point for model (2.4) is $\left(\frac{\Lambda_h}{\mu_h}, 0, \frac{\Lambda_v}{\mu_v}, 0\right)$. Moreover, $S_h = N_h$ and $S_v = N_v$.

Computing the jacobian at the DFE becomes,

$$J = \begin{pmatrix} -\mu_h & 0 & 0 & -\beta_{hv} \\ 0 & -(\gamma_h + \mu_h) & 0 & \beta_{hv} \\ 0 & -\beta_{vh}\dfrac{N_v}{N_h} & -\mu_v & 0 \\ 0 & \beta_{vh}\dfrac{N_v}{N_h} & 0 & -\mu_v \end{pmatrix} \tag{A.2}$$

Here, $N_h = \frac{\Lambda_h}{\mu_h}$, $N_v = \frac{\Lambda_v}{\mu_v}$. Consider $|J - \lambda I| = 0$ (I is identity matrix of same order). Expanding the matrix with respect to first column, and find an eigenvalue $\lambda_1 = -\mu_h$. The remaining matrix can be expanded in terms of second column, and we will find another eigenvalue $\lambda_2 = -\mu_v$. From the following matrix, the remaining two eigenvalues can be computed.

$$J_1 = \begin{pmatrix} -(\gamma_h + \mu_h) & \beta_{hv} \\ \beta_{vh}\dfrac{N_v}{N_h} & -\mu_v \end{pmatrix}$$

This matrix has eigenvalues with negative real part if and only if it contains positive determinant and its trace is negative. We have,

$$\text{Trace}(J_1) = -\mu_v - (\gamma_h + \mu_h) < 0.$$

Thus, trace of the matrix J_1 is surely negative. Now,

$$\text{Det}(J_1) = \mu_v(\gamma_h + \mu_h) - \beta_{hv}\beta_{vh}\frac{N_v}{N_h}.$$

Based on the condition that the determinant has to be positive implies,

$$\text{Det}(J_1) > 0 \Longrightarrow \mu_v(\gamma_h + \mu_h) - \beta_{hv}\beta_{vh}\frac{N_v}{N_h} > 0 \Longrightarrow \frac{\beta_{hv}\beta_{vh}\frac{N_v}{N_h}}{\mu_v(\gamma_h + \mu_h)} < 1.$$

That condition can be rewritten as $\mathfrak{R}_{00} < 1$, where the basic reproduction number is given by,

$$\mathfrak{R}_{00} = \frac{\beta_{hv}\beta_{vh}\frac{N_v}{N_h}}{\mu_v(\gamma_h + \mu_h)}.$$

In the vector-borne disease transmission process, there involve two stages:

- Host to vector transmission, and
- Vector to host transmission.

Reproduction number is the reflection of the average number of secondary infection individuals for each of the transmission stages, and the initial infection individual can be a human or a vector. Thus, the basic reproduction number $\mathfrak{R}_0$ and $\mathfrak{R}_{00}$ are equivalent.

Hence, $\mathfrak{R}_{00}$ is an equivalent form in terms of the threshold of $\mathfrak{R}_0$, and it can be presented as the product of the transmission from host to vector and vector to host:

$$\mathfrak{R}_{00} = \left(\frac{\beta_{hv}\frac{N_v}{N_h}}{\gamma_h + \mu_h}\right)\left(\frac{\beta_{vh}}{\mu_v}\right).$$

When $\mathfrak{R}_{00} < 1$, the system is asymptotically stable and the disease-free equilibrium point exists. The endemic equilibrium will exist when $\mathfrak{R}_{00} > 1$, and the disease may spread out through the population [1].

A.3. Endemic equilibrium

In order to find equilibria (endemic equilibria) of the system (2.4) where at least one of the infected components of the system (2.4) is non-zero, we need to take the following steps. To find the endemic equilibrium state of the model we set,

$$\frac{dS_h}{dt} = 0, \quad \frac{dI_h}{dt} = 0, \quad \frac{dS_v}{dt} = 0, \quad \frac{dI_v}{dt} = 0.$$

Suppose, $E^* = (S_h^*, I_h^*, S_v^*, I_v^*)$ represents any arbitrary endemic equilibrium of the model (2.4). Thus, system (2.4) can be written as,

$$\begin{cases} \Lambda_h - \beta_{hv} I_v^* \frac{S_h^*}{N_h} - \mu_h S_h^* = 0, \\ \beta_{hv} I_v^* \frac{S_h^*}{N_h} - (\gamma_h + \mu_h) I_h^* = 0, \\ \Lambda_v - \beta_{vh} S_v^* \frac{I_h^*}{N_h} - \mu_v S_v^* = 0, \\ \beta_{vh} S_v^* \frac{I_h^*}{N_h} - \mu_v I_v^* = 0. \end{cases} \tag{A.3}$$

Solving the equations of the above system at steady state gives,

$$S_h^* = \frac{\Lambda_h N_h Q_1}{\mu_h N_h Q_1 + \beta_{hv} Q_2 I_h^*}, \quad S_v^* = \frac{N_h \Lambda_v}{\mu_v N_h + \beta_{vh} I_h^*},$$

$$I_v^* = \frac{Q_2 I_h^*}{N_h \mu_v^2 + \mu_v \beta_{vh} I_h^*}$$

Let,

$$\begin{aligned} Q_1 &= N_h \mu_v^2 + \mu_v \beta_{vh} I_h^* \\ Q_2 &= \Lambda_v \beta_{vh} \\ Q_3 &= \gamma_h + \mu_h \\ Q_4 &= \beta_{hv}\beta_{vh} \\ Q_5 &= \mu_h \mu_v \end{aligned}$$

If $I_h^* \neq 0$, then substituting S_h^*, I_v^* in the second equation of the system (A.3) at steady state, we obtain after some calculations the following quadratic equation:

$$f(I_h^*) = P_1 I_h^{*2} + P_2 I_h^* + P_3 = 0 \tag{A.4}$$

where,

$$\begin{aligned} P_1 &= \mu_v^2 Q_3 \mu_h \beta_{vh}^2 + \mu_v Q_2 Q_3 \frac{Q_4}{N_h} \\ P_2 &= Q_3 Q_5 \mu_v^2 \beta_{vh} N_h + \mu_v^2 Q_3 \Lambda_v Q_4 + \mu_v^2 Q_3 Q_5 \beta_{vh} N_h - \frac{Q_2 Q_4}{N_h} \Lambda_h \mu_h \\ P_3 &= Q_3 Q_5 \mu_v^3 N_h^2 (1 - \mathfrak{R}_{00}) \end{aligned}$$

Clearly, the coefficient P_1 is always positive, and P_3 is positive (negative) if $\mathfrak{R}_{00}$ is less than (greater than) unity, respectively. Since $P_1 > 0$, the existence of the positive solutions of Eq. (A.4) will depend on the signs of P_2 and P_3. If $\mathfrak{R}_{00} > 1$, then there are two roots of Eq. (A.4) of which one root is positive and thus there is a unique endemic equilibrium. If $\mathfrak{R}_0 = 1$, then $P_3 = 0$ and there is a unique nonzero solution of Eq. (A.4), $I_h^* = -\frac{P_2}{P_1}$, which is positive if and only if $P_2 < 0$. If $P_2 < 0$ there is a positive endemic equilibrium for $\mathfrak{R}_{00} = 1$. Since equilibria depend continuously on $\mathfrak{R}_{00}$ which shows that there exists an interval of $\mathfrak{R}_{00}$ on which there are two positive equilibria,

$$I_{h1}^* = \frac{-P_2 - \sqrt{P_2^2 - 4P_1P_3}}{2P_1}, \qquad I_{h2}^* = \frac{-P_2 + \sqrt{P_2^2 - 4P_1P_3}}{2P_1}.$$

If $P_3 > 0$ and either $P_2 \geq 0$ or $P_2^2 < 4P_1P_3$, there are no positive solutions of Eq. (A.4), and thus there are no endemic equilibria. For the different ranges of these parameters, the following results are established [41].

Theorem 2. *The model (2.1) and (2.2) has:*

1. *a unique endemic equilibrium in Φ_1, Φ_2 if $P_3 < 0$ which is equivalent to $\mathfrak{R}_{00} > 1$,*
2. *a unique endemic equilibrium in Φ_1, Φ_2 if $P_2 < 0$, and $P_3 = 0$ or $P_2^2 - 4P_1P_3 = 0$,*
3. *two endemic equilibria in Φ_1, Φ_2 if $P_3 > 0$, $P_2 < 0$ and $P_2^2 - 4P_1P_3 > 0$, and*
4. *no endemic equilibria otherwise.*

Proof. An analogous proof can be found in [41], Theorem 4.1. □

Appendix B. Stability analysis

B.1. The local stability analysis of DFE

In this part, we investigate the local asymptotic stability (LAS) of the disease-free equilibrium point.

Theorem 3. *The DFE point of system (2.4) is LAS whenever $\mathfrak{R}_{00} < 1$ and is unstable if $\mathfrak{R}_{00} > 1$.*

Proof. The jacobian matrix evaluated at DFE point of the system (2.4) is,

$$J(P_0) = \begin{pmatrix} -\mu_h & 0 & 0 & -\beta_{hv} \\ 0 & -(\gamma_h+\mu_h) & 0 & \beta_{hv} \\ 0 & -\beta_{vh}\frac{N_v}{N_h} & -\mu_v & 0 \\ 0 & \beta_{vh}\frac{N_v}{N_h} & 0 & -\mu_v \end{pmatrix} \tag{B.1}$$

$J(P_0)$ has the following characteristic equation,

$$|J(P_0) - \lambda I| = 0. \tag{B.2}$$

where, I becomes the identity matrix. Hence, we get the polynomial in λ, say,

$$(\lambda+\mu_h)(\lambda+\mu_v)\left[-\frac{N_v}{N_h}\beta_{hv}\beta_{vh} + (\lambda+\gamma_h+\mu_h)(\lambda+\mu_v)\right] = 0 \tag{B.3}$$

The two eigenvalues of the characteristic equation are given by $\lambda = -\mu_h < 0$, and $\lambda = -\mu_v < 0$. The remaining two eigenvalues of the characteristic equation are given by the equation,

$$-\frac{N_v}{N_h}\beta_{hv}\beta_{vh} + (\lambda+\gamma_h+\mu_h)(\lambda+\mu_v) = 0. \tag{B.4}$$

After simplifying the above equation, we get,

$$\lambda^2 + c_1\lambda + c_2 = 0. \tag{B.5}$$

where,

$$\begin{aligned} c_1 =& \gamma_h + \mu_h + \mu_v \\ c_2 =& \mu_v(\gamma_h+\mu_h) - \beta_{hv}\beta_{vh}\frac{N_v}{N_h} \\ =& \mu_v(\gamma_h+\mu_h)(1-\mathfrak{R}_{00}). \end{aligned} \tag{B.6}$$

Clearly $c_1 > 0$ but $c_2 > 0$ if and only if $\mathfrak{R}_{00} < 1$. Hence, P_0 is LAS (local asymptotic stability) if $\mathfrak{R}_{00} < 1$, and unstable if $\mathfrak{R}_{00} < 1$ according to Hurwitz–Routh criterion [42]. □

B.2. The global stability analysis of DFE point's

We analyze the global asymptotic stability (GAS) of P_0 based on the value of $\mathfrak{R}_{00}$ using the conditions in the Castillo-Chavez method [43] in this section. To prove the GAS of P_0, the feasible region $\mathbb{C}_1 = \{(S_h, I_h, S_v, I_v) \in \mathbb{C}_1 : S_h \le S_h^0, S_v \le S_v^0\}$ is used, where $S_h^0 = \frac{\Lambda_h}{\mu_h}$ and $S_v^0 = \frac{\Lambda_v}{\mu_v}$. Firstly, region $\mathbb{C}_1$'s positively invariant property is established [44].

Theorem 4. *For the vector-host model (2.4), the region $\mathbb{C}_1 = \{(S_h, I_h, S_v, I_v) \in \mathbb{C}_1 : S_h \le S_h^0, S_v \le S_v^0\}$ is a positively invariant set.*

Proof. From system (2.4), we obtain,

$$\begin{aligned} \frac{dS_h}{dt} =& \Lambda_h - \beta_{hv}I_v\frac{S_h}{N_h} - \mu_h S_h \\ \le& \Lambda_h - \mu_h S_h. \end{aligned}$$

Hence we get,

$$S_h(t) \le S_h^0 - (S_h^0 - S_h(0))e^{-\mu_h t}. \tag{B.7}$$

If $S_h(0) \in \mathbb{C}_1$, then $S_h(0) \le S_h^0$. Therefore, $S_h(t) \le S_h^0 - (S_h^0 - S_h(0))e^{-\mu_h t} \le S_h^0$, for all $t \ge 0$. Similarly, we deduce from the third equation of the system (2.4), $S_v(t) \le S_v^0$, for all $t \ge 0$ if the initial conditions $S_h(0) \in \mathbb{C}_1$ and $S_v(0) \in \mathbb{C}_1$, respectively. As a result, region $\mathbb{C}_1$ is a positively invariant set that attracts all of the system (2.4) solutions in $\mathbb{R}_+^4$. □

Theorem 5. *For system (2.4), the disease-free equilibrium point P_0 is globally asymptotically stable in the region $\mathbb{C}_1 = \{(S_h, I_h, S_v, I_v) \in \mathbb{C}_1 : S_h \le S_h^0, S_v \le S_v^0\}$ if $\mathfrak{R}_{00} < 1$.*

Proof. The system (2.4) is written as,

$$\frac{dZ}{dt} = \mathbb{K}(Z, \mathcal{D}) \tag{B.8}$$

$$\frac{d\mathcal{D}}{dt} = \mathbb{L}(Z, \mathcal{D}) \tag{B.9}$$

where, $Z = (S_h, S_v)^T$ and $\mathcal{D} = (I_h, I_v)^T$.

$$\mathbb{K}(Z,\mathcal{D}) = \begin{pmatrix} \Lambda_h - \beta_{hv}I_v\frac{S_h}{N_h} - \mu_h S_h \\ \Lambda_v - \beta_{vh}S_v\frac{I_h}{N_h} - \mu_v S_v \end{pmatrix},$$

$$\mathbb{L}(Z,\mathcal{D}) = \begin{pmatrix} \beta_{hv}I_v\frac{S_h}{N_h} - (\gamma_h+\mu_h)I_h \\ \beta_{vh}S_v\frac{I_h}{N_h} - \mu_v I_v \end{pmatrix}$$

Here, $Z \in \mathbb{R}^2$ denotes the number of uninfected individuals, and $\mathcal{D} \in \mathbb{R}^2$ represents the number of infected populations including latent, infectious, and others. Furthermore,

$$A = \begin{pmatrix} -(\gamma_h+\mu_h) & \beta_{hv}\frac{S_h^0}{N_h} \\ \beta_{vh}\frac{S_v^0}{N_h} & -\mu_v \end{pmatrix}, \qquad \widehat{\mathbb{L}}(Z,\mathcal{D}) = \begin{pmatrix} \frac{\beta_{hv}}{N_h}(S_h^0 - S_h)I_v \\ \frac{\beta_{vh}}{N_h}(S_v^0 - S_v)I_h \end{pmatrix}$$

At $\mathcal{D} = 0$, $\mathbb{K}(Z,\mathcal{D})$ and $\mathbb{L}(Z,\mathcal{D})$ become,

$$\mathbb{K}(Z,\mathcal{D}) = \begin{pmatrix} \Lambda_h - \mu_h S_h \\ \Lambda_v - \mu_v S_v \end{pmatrix}, \qquad \mathbb{L}(Z,\mathcal{D}) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

where, $A = D_I\mathbb{L}(Z^*, 0)$, $\mathbb{L}(Z,\mathcal{D}) = A\mathcal{D} - \widehat{\mathbb{L}}(Z,\mathcal{D})$ and $Z^* = (\frac{\Lambda_h}{\mu_h}, \frac{\Lambda_v}{\mu_v})$.

Solving the system $\frac{dZ}{dt} = \mathbb{K}(Z, 0)$, we get $S_h(t) = \frac{\Lambda_h}{\mu_h} + (S_h(0) - \frac{\Lambda_h}{\mu_h})e^{-\mu_h t}$, and hence $\lim_{t\to\infty} S_h(t) = \frac{\Lambda_h}{\mu_h} = S_h^0$ and similarly from the other equation of $\frac{dZ}{dt} = \mathbb{K}(Z, 0)$, we get $\lim_{t\to\infty} S_v(t) = \frac{\Lambda_v}{\mu_v} = S_v^0$. The solutions of $\frac{dZ}{dt} = \mathbb{K}(Z, 0)$ do not depend on the initial conditions $S_h(0)$ and $S_v(0)$. As a result, the asymptotic nature of the solutions $S_h(t)$ and $S_v(t)$ are independent of the preliminary conditions in $\mathbb{C}_1$, ensuring the global asymptotic stability of the equilibrium point, $Z^* = (\frac{\Lambda_h}{\mu_h}, \frac{\Lambda_v}{\mu_v})$ and hence the first condition of the Castillo-Chavez method [43] is satisfied. Further in $\mathbb{C}_1$, $S_h \le S_h^0, S_v \le S_v^0$ and hence,

$$\frac{\beta_{hv}}{N_h}(S_h^0 - S_h) \ge 0,$$

$$\frac{\beta_{vh}}{N_h}(S_v^0 - S_v) \ge 0.$$

This implies that

$$\widehat{\mathbb{L}}(Z,\mathcal{D}) \ge 0,$$

which satisfies the second condition of the Castillo-Chavez method [43]. As a result, P_0 is GAS (globally asymptotically stable) if $\mathfrak{R}_{00} < 1$. □

Appendix C. Sample codes

```
function dy = malaria(t,w)
global lamda1 lamda2 muh muv gamma b1 b2 n
sh = w(1); ih = w(2); rh = w(3); sv = w(4); iv= w(5);
dy(1)=lamda1-(b1*iv*sh)/n-muh*sh;
dy(2)=(b1*iv*sh)/n-(gamma+muh)*ih;
```

```
dy(3)=gamma*ih-muh*rh;
dy(4)=lamda2-b2*sv*(ih/n)-muv*sv;
dy(5)=b2*sv*(ih/n)-muv*iv;
dy = [dy(1);dy(2);dy(3);dy(4);dy(5)];

clear all
clc
global lamda1 lamda2 muh muv gamma b1 b2 n
t0 = 0; tfinal = 1000;
lamda1=0.000039;
lamda2=0.143;
muh=0.00039;
muv=0.1;
gamma=0.2;
b1=0.225;
b2=0.375; n=100; sh0 = 99;
ih0 = 1;
rh0 = 0;
sv0 = 999;
iv0 = 1;
w_ini = [sh0, ih0, rh0, sv0, iv0];
[t, w] = ode45('malaria', [t0, tfinal], w_ini);
lablevec = {'S_h','I_h','R_h','S_v','I_v'};
plot(t,w(:,1),'r',t,w(:,2),'–b',t,w(:,3),'-*g','LineWidth',4)
box off; hold on; set(gca,'linewidth',2)
xlim([0 60]);title('Determinictic SIR for host','FontWeight','bold',
'FontSize',17);
xlabel('Time','FontWeight','bold','FontSize',15);
ylabel('Population','FontWeight','bold','FontSize',15);
set(gcf, 'PaperPositionMode', 'manual');
set(gcf, 'PaperUnits', 'inches');
h2 = legend('S_h','I_h','R_h','FontWeight','bold','FontSize',22,'Location',
'east','Orientation','Vertical');
legend(h2,'show')
legend boxoff
```

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